

Personalized Educational Game in Human-Agent Systems

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Abstract

Due to technology's rapid development, people live under the growing impact of IoT devices. While the IoT ecosystem could bring convenience, the potential privacy threats are also non-negligible, especially for leaking sensitive personal data. There have been many related works focusing on solving privacy issues in the area of mobile apps and IoT devices. In this project, we designed an educational game to help people better understand their privacy preferences and learn more about IoT devices. Following the ethics rules, we conducted a user study to evaluate the game and the system's usability. The results indicate that our project was successful and achieved our aims and educational tasks. This project also provides insights to design a better system for analyzing privacy preferences and educational purposes in the future.

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Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Yibin Xia)

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Chapter 1

Introduction

1.1 Motivation

Due to the rapid development of technology in recent years, there has been a trend that more and more people live under the influence of IoT devices [10]. As its name suggests, the Internet of Things (IoT) is a vast network that connects various kinds of information sensing and collecting devices to build interconnections between humans, intelligent agents, and edge devices [6]. Invented by Bill Gates, the concept of IoT has been mentioned long since 1995. While in these days, IoT devices such as air quality monitors, voice controllers, smart speakers, etc., are gradually involved in people's daily lives and have become the most popular intelligent agents [3].

While the IoT ecosystem could provide convenience to people, it could also bring us privacy issues. For example, a temperature sensor equipped in the kitchen may measure the room temperature, a facial recognition-based payment system equipped in a department store may scan and store customer's facial information, a GPS tracker in a smartwatch may read user's local position. Subsequently, the data collected by the connected IoT device may be shared with other authorities or uploaded to the cloud server without user's awareness. More importantly, the data shared with others are usually sensitive, which generally involves our personal or biometric information. The authorities who collect our data have different purposes. They may use the data for cloud-based big data analysis or do improper things such as illegally leaking the data to third parties without notifying us and such that our privacy rights would be threatened [16].

In the research of Helen Nissenbaum et al., they pointed out that privacy issues have been one of the most prevalent social issues in the area of digital electronic information

technologies, and researchers have concluded that privacy issues are complex and hard to conceptualize [21, 26]. One of the main reasons behind this is that people usually don't have their privacy preferences in mind, such that they don't know whether to allow or deny when they are facing data collection requested by IoT devices [1]. However, the privacy preference in the IoT ecosystem are complicated because they are changeable under different scenarios. People will face different data collection requests in various situations, which are composed of several factors such as specific location, what kind of data will be collected, the purpose of gathering that data, etc, and those factors are usually complicated and interrelated. Consequently, it would be better for people to learn much more about their privacy preferences when deal with IoT devices.

1.2 Project Aim

To relieve the privacy issues mentioned in the previous section, the first goal of this project aims to develop a personalized educational game for players that can help them better understand their privacy preferences by providing them with customized feedback according to their reactions. Players will face different scenarios simulated in the game, and they are asked to provide reactions to each scenario. The game is implemented with a preference analysis model, which collects players' reactions to each scenario and generates preference analyzing results for them when they have finished the game.

The other goal of this project is to combine studying with fun. Since this is an educational game, we are focusing on how to have fun while study. In addition to the privacy preference analysis model, the game will examine players' basic knowledge about IoT devices through several pop-up quizzes interspersed in each scenario and provide feedback for their performance on the examinations.

A user study was conducted for the evaluation of this game. The user study includes a game experience questionnaire that evaluates the overall design and gameplay of the educational game, and a form of system usability scale (SUS) to assess the usability and learnability of the system, as well as a summative comment regarding players' privacy preferences.

1.3 Report Structure

The structure of this dissertation is organized as follows:

Chapter 2 will provide some background knowledge and related work about this project which will help better understand the details of the game content and methodology used to implement the game.

Chapter 3 will illustrate and explain the methodology used for the game, including the underlying logic of how each game level was designed, the algorithm we applied for preference analysis model, and the arrangement of each scenario for privacy questions.

Chapter 4 will firstly introduce the overall design of the game, including the flowchart and the main page interface, and then the design details for each part of the game will be described respectively.

Chapter 5 will dive into the implementations of the game in detailed level, and some techniques with Unity and C# scripts will also be introduced.

Chapter 6 will be the evaluation and user study of this project, which includes a game experience survey and a further study on analyzing privacy preferences.

Chapter 7 will recapitulate what we have done in this project and make a summation for our project.

Chapter 2

Backgrounds

This chapter will explore and briefly summarize some background knowledge and related work of this project. Specifically, we will firstly introduce and give some examples of cases that incorporate video games with studies, and then we will explore some factors which could affect people's preference, and explain some knowledge behind the algorithm we applied in the analysis of privacy preference.

2.1 Educational Games

When referring to video games, the conventional mindset tells us that video games are usually for recreational purposes and have negative impact on studies if addicted to games. However, we would also benefit from video games if we take advantage of them appropriately. Some studies have shown the benefits of incorporating video games into the study environment, illustrating the improvements in problem-solving skills, learning interest, and knowledge acquisition motivation. [5, 7, 23]. For example, in the research of Hyungsung et al., a meta-cognitive strategy designed in game-based learning was proved that can effectively increase student's learning performance [7]. In addition, another game-based learning system developed by Chung-Ho Su et al. also shows that compared with traditional face-to-face teaching, game-based learning could help students gain more learning motivations. [5].

Other than the games-based study environment investigated by researchers, there are also many educational games developed by game companies. For a crazy fan of astronomy and astrophysics, he or she must have heard **Universal Sandbox**, an excellent space simulator equipped with a physics engine that can simulate the real universe and perform any astronomical phenomenon [15]. As one of the most representative edu-

cational games, many people have acquired much knowledge about the universe and astrophysics through **Universal Sandbox**. Another famous example of educational game is civilization series. First released in 1991, the civilization series had sold over 33 million copies and attracted more than forty thousand players who are in love with history, politics, and economics [17].

2.2 Factors Impacting Privacy Preference

In a recent study by Almuhimedi et al., they have an inspirational discovery that people's privacy preferences could be more effectively controlled by using a permission manager app that takes records of users' choices each time they encounter data collection scenarios by IoT devices [2]. This study gives us an insight that if we could provide information about what factors could affect people's privacy preferences, we could help them better control their privacy issues and avoid potential privacy threatens.

In the context of mobile devices and applications, many studies focus on analyzing factors that could impact individuals' privacy expectations by analyzing their comfort level and willingness to share data [14, 25, 29]. For example, Sadel et al. built a system for users of mobile devices to manage the permissions for sharing users' locations with others, and they pointed out that it is crucial to improve users' awareness in order to protect their privacy [25]. Additionally, Tsai et al. investigated factors about the feedback of data being shared, and they concluded that people are more comfortable when they receive adequate feedback [29]. Furthermore, according to the research conducted by Lin et al., the purpose for which an app requests access to sensitive data and the individual expectations about the behaviors of the app could influence individuals' privacy decisions [14].

The studies about factors impacting privacy preference are not limited to mobile apps. In the aspect of IoT data collection, Lee and Kobsa conducted two separate studies investigating five factors of privacy data collection, and they concluded that individuals' privacy preferences are highly dependent on the type of IoT devices, such as monitors or cameras. Their studies also indicated that people would feel much uncomfortable if they are monitored by unknown entities or the government [12, 13]. In addition to the effect of device type, there are two more factors being studied by Lederer et al.: the situation in which the data is being collected and the entity that manages the data. Their results showed that the relative importance of who is collecting the data is much greater than the importance of the situation of data collection [11]. Based on

the previous studies mentioned above, Naeini et al. did research on predicting privacy expectations by conducting a within-subjects survey which collected the privacy preferences of more than one thousand participants on different scenarios. They also used machine learning methods to train a model that can predict people's privacy expectations, and the model achieved an average accuracy of over 80% on predicting comfort level and over 75% on predicting allow/deny decisions [19].

2.3 Statistical Correlation Coefficients

Similar to the work of Naeini et al., a preference analysis model will be implemented in this project. The model will take factors that impact preference analysis as input and outputs relative importance coefficients for each factor. The group of Naeini et al. construct several generalized linear mixed models (GLMM) which are particularly useful for modeling experiments with multiple parallel scenarios [19]. Besides that, many other machine learning models such as support vector machine (SVM) and logistic regression model are also applicable in this project [20, 27].

Even though machine learning models are powerful, they cost time for training and data preprocessing, and they also need a large number of the dataset to improve accuracy. Hence it would be handy if typical statistical methods were used. There are three kinds of statistical correlation coefficients usually used to represent the significance levels between each random variable: Kendall rank correlation coefficient (KRCC), Spearman's rank correlation coefficient (SRCC) and Pearson product-moment correlation coefficient (PPMCC). The PPMCC, also known as simple correlation, calculates the coefficients of linear relationship between two random variables [8], while the SRCC, as its name suggests, measures the coefficients of ordinal random variables which are nonlinear [24]. Both SRCC and KRCC could be used for discrete values, but the only difference between them is that KRCC is not suitable for ordinal variables. Ordinal represents that there is a ranking between random variables such as score ranking, population ranking etc. The value of each random variable doesn't matter its coefficient, but the relative order between each random variable does matter. We adopted SRCC in this project and the specific algorithm will be introduced in chapter 3.

Chapter 3

Methodology

In this chapter, we will cover the underlying logic of designing each game level, the arrangement of each scenario for privacy questions, and the algorithm we chose to analyse privacy preference.

3.1 Code Book of Factors Impacting Preferences

Naeini et al. investigated the importance of eight factors when presented to individuals in combination, and they ranked those factors based on corresponding Bayesian Information Criterion (BIC) values [19]. The lower the BIC values, the higher the significance level for a particular factor. According to those BIC values, we picked up six factors that have the highest significance level:

- Location: Where the data is collected
- Device type: What kind of IoT devices cast the data collection request
- Purpose: The purpose why the data is collected
- Retention time: How long the data will be stored
- Shared: Whether the data will be shared with third parties
- Benefit: Who will benefit from the data collection

The code book for each factor is shown in Table 3.1, which lists the name for those six factors, the code values for different corresponding levels, and the ranking criterion on which to rank those levels. To fit the data into the preference analysis model, we carefully assigned a value for each factor based on different criterion such that the

Spearman's rank correlation coefficient could accurately represents the correlation between each factor. The values range from 0 to 1 to omit normalization processes in order to increase calculation speed.

The factor **Location** is divided into six levels based on the privacy confidentiality of each space. The highest space confidentiality will achieve the highest score and vice versa. We consider that *your home* has the highest space confidentiality, and then *your friend's home*, and so on. This is a quite straightforward and intuitive way to rank the level of confidentiality. We combined the factor **Device type** and **Data type** as Naeini et al. did in their research [19] since these two factors are highly correlated. For any specific IoT device, it usually collects a certain type of data. For instance, iris scanner will always collect people's iris information almost without any exception. We evaluated the sensitivity of data collected by IoT devices and assigned a larger value to the device that collects more sensitive data. Those devices that collect biometric data, such as fingerprint scanners and facial recognition systems, were given higher values. In comparison, lower values were given to those devices which detect environmental information, such as temperature sensors.

There are more than thirteen different **purposes**, and for each specific purpose, the value will be evaluated individually according to the context. Some of the purposes are intended to help data collectors fulfil other aims irrelevant to users, while others intend to provide a better user experience. In general, we will assign lower values to the former and higher values to the latter. There are some cases where data collectors did not provide any reason why they requested to collect data, in which cases values of 0 will be assigned. The ranking criteria for **Retention time** is quite straightforward. A longer retention time corresponds to a higher value and vice versa. Notably, the value of *unspecified* is lower than that of *not deleted* because we considered that unspecified data retention time is assumed to be a period longer than one year but not forever. For the rest of the two factors, **Being shared** and **Benefit**, their codes of levels are binary values; hence either 0 or 1 will be assigned.

3.2 Scenario Arrangement

As we mentioned in the project aim, one of the goals of this project is to analyse individual's privacy preference in different scenarios. The data collection scenario in this project consists of combinations of six factors listed in Table 3.1, where the factor

Factor name	Levels(code)	Count	Ranking criteria
Location	1. Department store (0)	15	Space confidentiality
	2. Work (0.2)	15	
	3. Library (0.4)	15	
	4. Public restroom (0.6)	15	
	5. Friend's house (0.8)	7	
	6. Home (1)	8	
Device type	1. Temperature sensor (0)	16	Sensitivity of data which was collected
	2. Presence sensor (0.2)	17	
	3. Smartwatch (0.5)	8	
	4. Smartphone (0.5)	6	
	5. Camera (0.7)	15	
	6. Facial recognition (0.9)	5	
	7. Fingerprint scanner (1)	5	
	8. Iris scanner (1)	3	
Purpose	A-M different specific purpose	N/A	Each specific purpose will be analyzed solely
Retention time	1. Purpose satisfied (0)	14	The length of time
	2. Week (0.2)	14	
	3. Year (0.4)	15	
	4. Unspecified (0.8)	16	
	5. Not deleted (1)	15	
Being shared	1. Shared(1)	24	Binary value
	2. Not shared(0)	51	
Benefit	1. User(1)	64	Binary value
	2. Data collector(0)	9	

Table 3.1: List of factors and the level code for different levels of factors, as well as the ranking criteria for which levels are ranked

levels varied. A simple permutation and combination calculation tells us that there are

$$6 \times 8 \times 13 \times 5 \times 2 \times 2 = 12480$$

possible combinations in a privacy scenario. Fortunately, in the research of Naeini et al. they have well-designed nearly 330 scenarios in which factors are combined in a rational and reasonable way [19]. Based on this scenario dataset, we carefully selected 75 scenarios with all levels of factors evenly distributed such that they share almost the same weight in calculating their correlated coefficients. The number of occurrences each level appeared in 75 scenarios is also showed in Table 3.1.

Note that the *smartwatch* and *smartphone* shared the same code value because they collect the same kind of data type, which is *specific position*. Consequently, these devices are considered identical when calculating the correlated coefficients. The same thing also applied to *fingerprint scanner* and *iris scanner* since they all collect biometric information.

3.3 Privacy Preference Analysis Model

When all factors are encoded into ordinal values using the code book in Section 3.1, we can use the model to calculate the significance levels for these factors. The significance levels are evaluated based on two criterion: decision and comfort level. During the game session, we will present 75 data collection scenarios to players and ask them some privacy questions, including how they feel about the data collection in the described situation and whether they allow or deny it. We then take records of players' responses and use the statistical models to evaluate the significance levels of factors corresponding to the responses.

Responses	Variable Type
Allow/Deny Decision	Binary: allow(1), deny(0)
Comfort Level	Ordinal: Very uncomfortable(1), Uncomfortable(2), Neither comfortable nor uncomfortable(3), Comfortable(4), Very comfortable(5)

Table 3.2: The variable type of responses to privacy questions

Table 3.2 shows the variable type of player's responses that we will record in the game. Decisions on data collections are indicated as binary outcome variables where

approving represents one and rejecting represents zero. The variable type for comfort level is also in an ordinal form on a scale from 1 to 5. It is worth noting that the rankings of decisions and comfort levels are in the same direction, which means the higher the rank, the more likely that players are happy with the data collection. For example, the highest ranking in comfort level is 5 (very comfortable). We followed the convention that if people are more comfortable with the data collection, they are more likely to approve the collection request. Although there might be some exceptions, it would be helpful when we present results to players.

We use Spearman's rank correlation coefficient (SRCC) to calculate the correlated coefficients. The formula of SRCC is showed in equation 3.1, among which d_i represents the rank difference between two random variables X_i and Y_i . It was proven that r_s ranges from -1 to 1, where the sign of r_s represents the monotonic relationship between dependent and independent variables, and the absolute value of r_s indicates the degree of monotonic relationship [18]. In other words, there will be a robust negative association between two variables if r_s approaches -1, a strong positive association if r_s approaches 1, and no apparent monotonic association if r_s approaches 0.

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (3.1)$$

3.4 Pop-up Quiz

To fulfil another goal of this project, we designed several pop-up quizzes to examine players' knowledge about IoT devices. There will be ten quizzes randomly distributed in 75 scenarios. The structure of the quizzes is as follows. Firstly, players will be presented in a simulated scenario, such as in a department store or a library, and then they are told that a kind of IoT device is trying to collect their data. Players will not be directly told what the device is, but they will be hinted at by side information such as the function or feature of that IoT device. Afterwards, five icons will be randomly generated through a device pool and players are required to select the correct icon that accurately represents the IoT device mentioned in the quiz. We have prepared eleven icons of IoT devices in the device pool: headphone, iris scanner, presence sensor, fingerprint scanner, facial recognition, temperature sensor, personal computer, camera, smartphone, and smartwatch, respectively.

Each quiz will focus only one IoT device. The number of quizzes for different type of IoT devices is showed in Table 3.3. The results control system implemented in

Device name	Count
Facial recognition	1
Presence sensor	3
Iris scanner	1
Fingerprint scanner	2
Camera	1
Temperature sensor	1
Smartphone	1
Total number of quizzes	10

Table 3.3: Number of quizzes subject to the type of IoT device

the game will records players answers to the quizzes and evaluate their overall performance and accuracy for specific devices to provide tailored feedback to them. More details about the design of results control system as well as the preference analysis model will be introduced in chapter 4.

Chapter 4

Game design

In this chapter, we will introduce this project's game design from the overall and detailed perspectives, respectively. The flowchart for the game and some screenshots of the interfaces will be presented.

4.1 Overall Design

The development tool we chose is Unity [28], a user-friendly 2D and 3D multi-platform game development engine. Unity has been one of the most popular development platforms, with countless free game materials online, including graphic user interfaces, background music and prefabs. When the game was designed initially, considering we needed to show players as many as 75 scenario-based privacy questions along with the pop-up quizzes, it would be extraordinarily time-consuming and impossible to finish in a timely scale if we built up the whole scene for each privacy question, either in 2D or 3D. Hence we changed the idea to build a text-based interactive game.

4.1.1 Flowchart

Figure 4.1 shows the flowchart of this game. When the game is started, there are two options: start a new game and load previous save data. We provided a save function that records the players' data so that if players want to suspend the game sessions, they can continue their sessions next time by loading the data files. A tutorial will be shown if it is the first time playing the game. Afterwards, players will proceed to the main page, which connects every game scene. As shown in the flowchart, the six blocks right below the main page block connected by bidirectional arrows represent six different

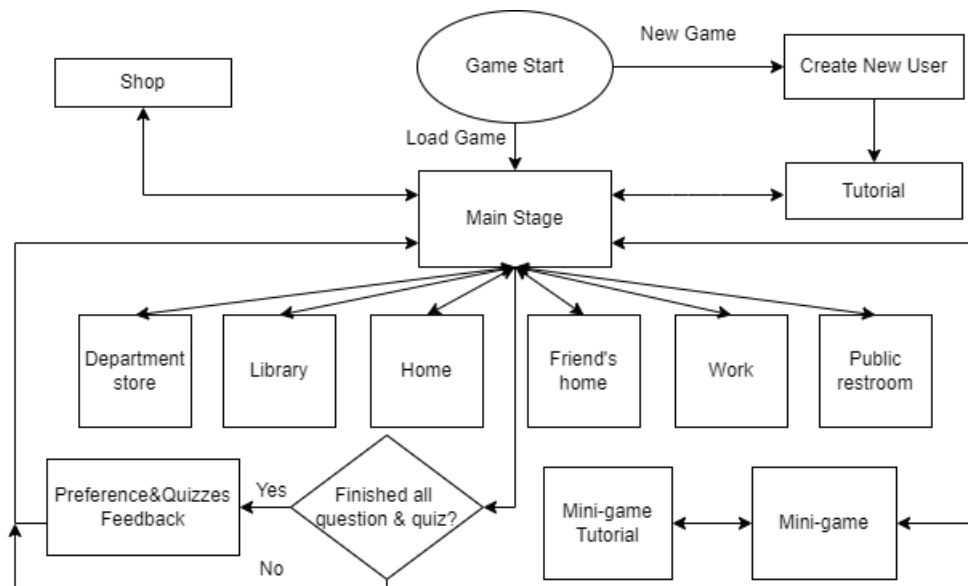


Figure 4.1: The flowchart of the game

scenario locations, which contain privacy questions and pop-up quizzes. Players can check their personalized feedback for privacy preferences and pop-up quizzes when they finish privacy questions for all six locations. The mini-game and shop scenes could also be accessed from the main page.

4.1.2 Main page

Figure 4.2 shows the layout of the main page. The left and right corners indicate players' username, current points earned through the mini-game, and the number of correct answers they got in the pop-up quizzes. The points are used to buy items in the shop, which is shown in the form of a shopping cart icon on the top right side of the interface. The tips icon, a yellow book with a light bulb, is a tutorial button where players could review the game's information and rules if they had skipped the tutorial before.

The middle six icons occupying more than half of the interface are buttons for different locations of privacy questions. Clicking one of the location buttons will direct players to a scene that will include every privacy event that happens at that location. For example, if players click the button tagged as department store, they will face several privacy questions related to the department store. Furthermore, the related IoT devices for pop-up quizzes will also appear in the department store. The reason why we did this is that we want to simulate scenarios as realistic as possible. When players finish

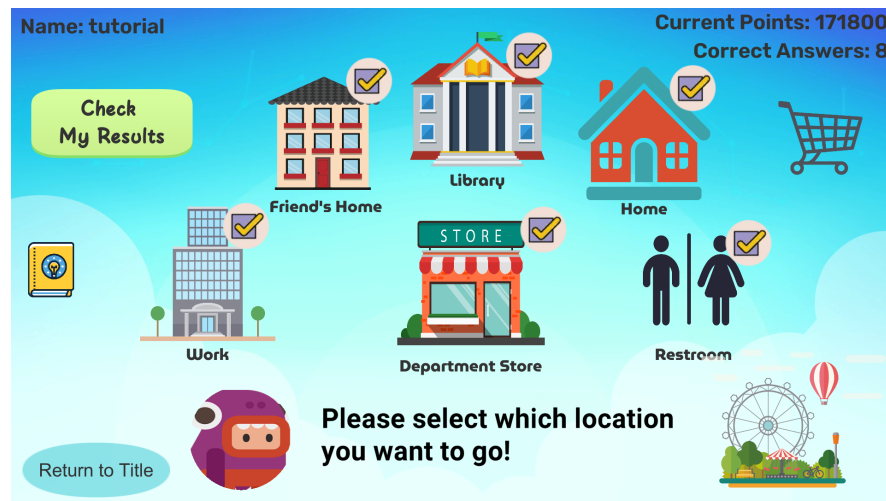


Figure 4.2: The interface of the main page

all privacy questions and pop-up quizzes on the department store, the corresponding icon will be ticked, and players can proceed to the following location. A green button on the top left side will be available to check their results when all six sites are ticked. The mini-game icon, located on the bottom right, provides an entry to the mini-game which will be introduced in the next section.

4.2 Design Details

In this section, we will introduce the design of this game in detailed level, which consists of several parts: introductory pages, scenario-based privacy questions, pop-up quizzes for IoT devices, mini-game, and results analysis model.

4.2.1 Introductory pages

When players first enter the game, they will be shown some introductory pages which will briefly introduce the game and help them to create their own save file. In the introductory pages, players are asked to input their username, as shown in Figure 4.3. The system will warn players if the inputted username is too long (over ten characters) or too short (below three characters). The system would also do a comparison search with the database to prevent name conflict. If a player inputs a username that other players already take, the system will also cast a warning text shown in Figure 4.3(c).

If the player inputs a valid username, a save file will be created automatically with some initialized data, and then the save file will be stored in the database. Players are

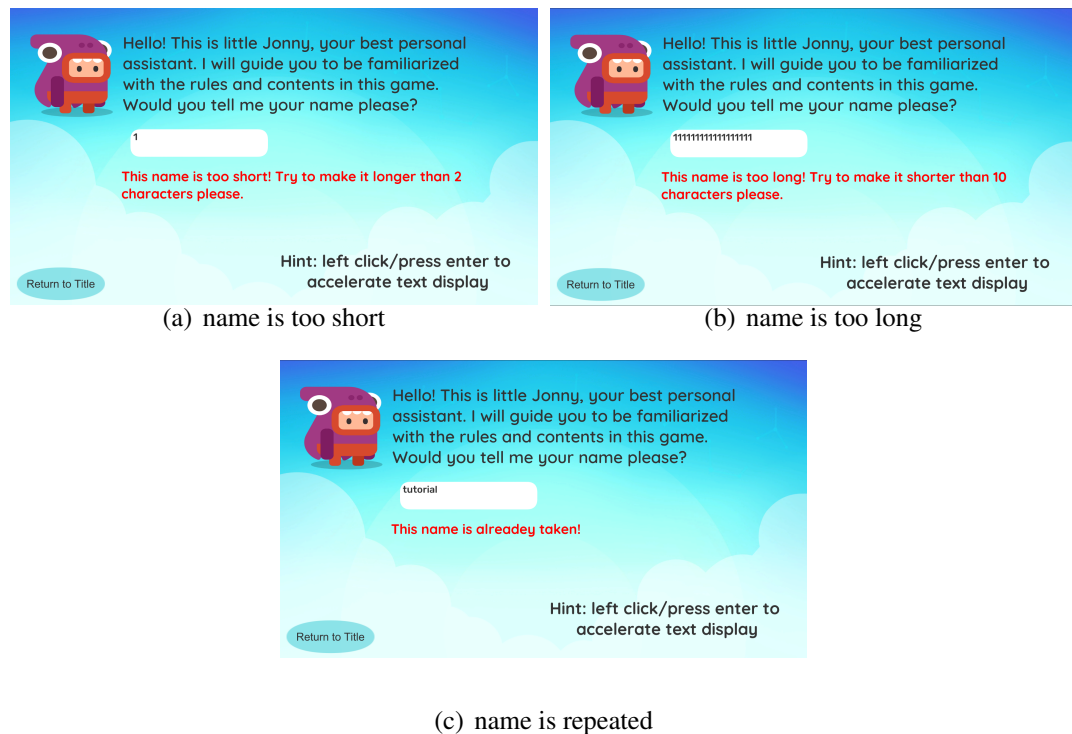


Figure 4.3: Introductory page: name input

initially given 15,000 points and two chances of tips for pop-up quizzes. They could get more points by playing the mini-game and use points to buy more chances of tips or other useful tools in the shop. Additionally, It is worth noting that there is a dialogue box on the bottom right of the introductory page, which tells players how to accelerate text display. We have made an animation that displays text character by character rather than showing everything at once. Players could control the display speed by left clicking or pressing the enter key. This design aims to improve interactivity with players.

4.2.2 Scenario-based privacy question

The 75 scenarios for privacy questions constitute the central part of this game. We will use a storytelling tone to vividly illustrate each scenario to players rather than mechanically state the whole question. Otherwise, the game would be tedious and concatenated by dullness. We also use second-person narration to simulate the scenario so players would feel like they are in the story. All the efforts on the narrative techniques aim to simulate a real life situation for players.

Figure 4.4 shows an example scenario of the interface for privacy questions. The



Figure 4.4: Interface of privacy question: Public Restroom

background is an image of the scene indicating the player's current location, which is the public restroom in this case. We use different colors to emphasize factors that may affect privacy preference, as shown in Table 4.1. By changing the colors, it would be helpful to present each scenario to players. Furthermore, not only using plain text, we also put some images of IoT devices in the background to illustrate the current type of device that collects privacy information. For example, there is a camera mounted on the ceiling and a presence sensor installed on the wall beside the restroom mirror. Another purpose of doing this is to imply to players the possible position where the IoT devices are installed in real life. Additionally, we added a reference list on both sides of the dialogue box so players could quickly check those factors on the privacy question.

Factor	Location	Device type	Data type	Retention	Shared	Purpose
Color	Red	Blue	Green	Purple	Magenta	Grey

Table 4.1: Different colors corresponding to different factors

Players are asked two privacy questions about their comfort levels and decisions regarding the scenario mentioned before when all factors are presented. The interface for selecting comfort levels and decisions is shown in Figure 4.5. The system will record players' choices as soon as they make their selections and store them in the database. Consequently, they do not have to save their data manually due to this autosave feature.

Another feature in the design of privacy questions is that we made each scenario



Figure 4.5: Interface of comfort level and allow/deny decision

consecutive, which means there is a relationship between two consecutive scenarios. For instance, a scenario would be a camera recording a video of the entire room in the public restroom, and the video will be shared with law enforcement. In this question, players are not told how long the data will be kept. Subsequently, we will keep every factor unchanged except for the retention time in the following scenario. Hence the next question would be how would they feel if players are told that the data will be kept for a certain amount of time, one year or until the purpose is satisfied. The reason behind this setting is that we considered that if every factor is changed, players would feel confused and could not make up their minds accordingly and resulting in a mess. The nuance of factors between two scenarios is helpful when analyzing the effect on individuals' privacy preferences.

4.2.3 Pop-up quizzes and tips

The pop-up quizzes usually appear before the privacy questions. The interface for pop-up quizzes is shown in Figure 4.6. Five icons, each representing one kind of IoT device, are located in the middle of the image. There will always be one correct answer, with the rest four icons randomly generated from the IoT device pool. A quiz tag resides on the top right corner of the interface, which indicates the current quiz ID and state. The quiz ID will be referenced on the quizzes' results page, which tells the player's performance on each quiz. Players could try the same quiz multiple times to improve their accuracy, and the number of tries will be recorded until they get correct. The quiz state will show whether players get the correct or wrong answer on that quiz, and show as none if it is the first time to attempt the quiz.

The tips icon on the right side of the quiz tag also appears along with the quiz. Players could consume one chance of tips to check the hints if they feel stuck on the



Figure 4.6: Interface of pop-up quizzes

quiz. There are two levels of tips. In the first level, we will hint to players by giving more information about the IoT device and providing some ideas to infer the correct answers. In the second level, we will directly tell players which icon is correct. The usage of tips will also be stored in the database to provide feedback about players' degree of knowledge on corresponding IoT devices. We also provide a shortcut to buy hints paying with points if players do not have one. The current number of chances and points will be presented if players decide to use hints.

4.2.4 Log and jump-scene button

Referring back to Figure 4.6, we could find a dialogue box in the top middle of the privacy question interface, which indicates the index of the current scenario over the total number of scenarios on this location. As mentioned in Section 4.1.2, a location icon includes a subset of scenarios whose privacy events happened in that location. The numbers of one subset of scenarios grouped by location are presented in Table 3.1 in the **Location** row. We could see that most locations have 15 scenarios except for *home* and *friend's home*, which are 7 and 8, respectively. We considered that 15 scenarios are too many for a single location, so we divided each location into several separate scenes, with each scene having 4 or 5 scenarios. When players finish the privacy questions in one scene, they can press the right arrow button to proceed to the next scene, which contains another 4 or 5 scenarios. A left arrow button also allows players to jump back to the previous scene. Furthermore, if players want to change their minds, instead of going through every scenario in normal order, they could use

this jump-scene button to jump to different scenes quickly.

To make the scenario texts self-contained, we added a log function that records the texts in the dialogue box and the player's choices. The log icon locates in the bottom right corner of the dialogue box. If players forget the context of the scenario or their choices, they could review it by clicking the log icon and then a scroll view will show up. Figure 4.7 shows the log of players' comfort level, decision and answer to a pop-up quiz, respectively. The logs will be cleared if players quit the game, move to another location, or jump to another scene.

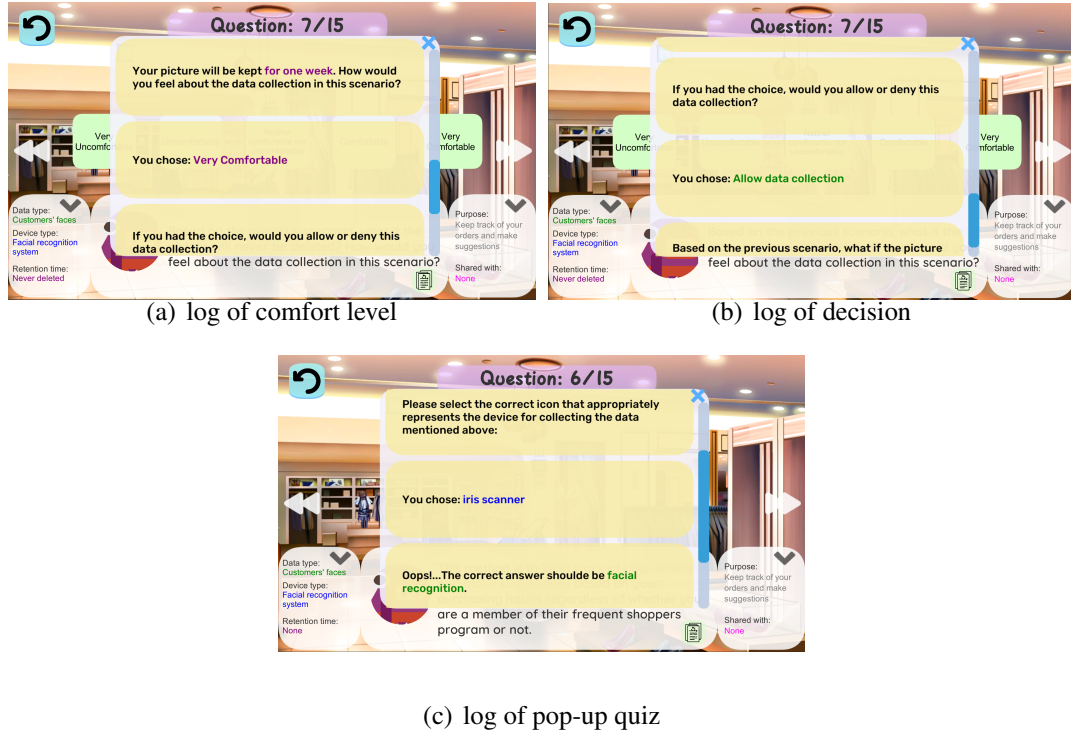


Figure 4.7: Logs for texts and player's choices

4.2.5 Mini-game and shop

In case players get bored when reading a large amount of text, we embedded a mini-game into this project. The name of this mini-game is **Happy Eliminating**, a great little game to pass the boring time. Due to its simple rule and unique playability, this game has been widely deployed on mobile devices, and we transferred it into our game. According to research about the playability of mobile games, in the evaluation of games, there are three essential modules: game usability, mobility, and gameplay [9]. We believe that deploying the game **Happy Eliminating** on our project would

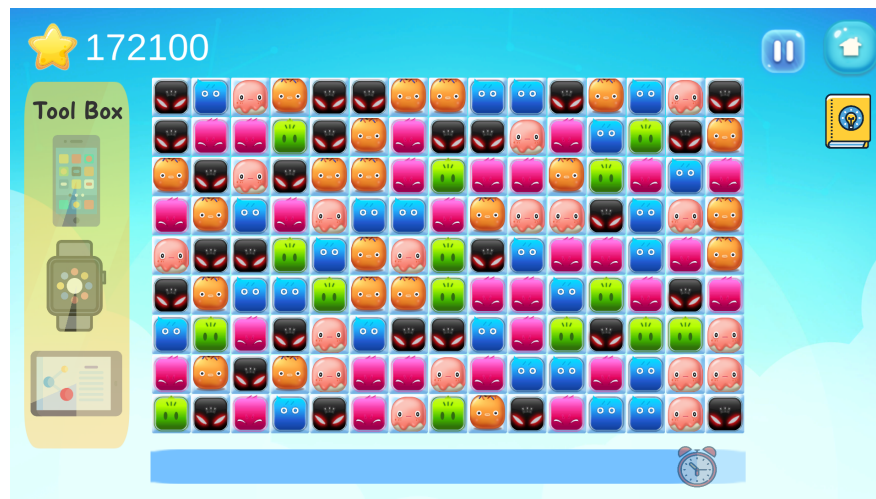


Figure 4.8: Interface of mini-game: Happy Eliminating

help improve the playability of our game.

The interface of the mini-game is shown in Figure 4.8. The players' current points are located in the top left corner, and the tutorials for the mini-game and other buttons reside in the top right corner. The game rule is simple, and players are easy to be addicted to it. There are nine rows and fifteen columns on the game board, with one of six different small blocks on each entry. The blocks are eliminated by putting more than three identical blocks on each entry. The blocks are eliminated by putting more than three identical blocks in a line, horizontally or vertically, but these blocks must be adjacent. New blocks will be generated at the top to substitute the eliminated blocks. Players could drag the block to switch its position with its adjacent block. If the requirements to eliminate blocks are not met after switching, the block would return to its original position. There is a timer right below the game board. We set the timer to 120 seconds, and the game is over when the timer counts down to zero.

A toolbox on the right side of the game board contains three kinds of IoT devices. These devices have various functions and could be bought from the shop. The first one is the smartphone, whose function is temporarily increasing the number of points for successfully eliminating each one of the blocks. It has a charge gauge and was initially disabled while eliminating blocks each time would charge the device. The device is ready to use when fully charged and needs to recharge after use. The exact mechanism also applies to the second IoT device, a smartwatch. The function of the smartwatch is to freeze the timer for a while. Different from the previous two devices, the third one, the tablet, does not need to charge. It is more potent in that it could randomly destroy one row of blocks, instantly giving a large amount of points, but it needs to cool down for a while after used.



Figure 4.9: Interface of the shop

The tools described above are available in the shop, whose interface is shown in Figure 4.9. Players could buy four items in the shop, three of which are IoT devices used in the mini-game, while the last item is tips for the pop-up quizzes. The points earned through the mini-game could be used to buy some IoT devices to get more points or buy tips to improve pop-up quizzes' accuracy. The cost for each item is located on the top right side of the product description. We would increase the cost of tips each time players buy one of them, which means it would be very costly if players bought too many tips. The costs for the other three items are fixed because there is only for each IoT device, there is only one left in stock.

4.2.6 Results pages

We divided the results pages into two parts. In the first part, we will present to players the most and the least factors that affect their privacy preferences. Additionally, we will use pie charts and bar charts from visualizing the six factors' correlation coefficients. Figure 4.10 is an example in the results pages of the visualization of factors that impact comfort level. Figure 4.10(a) is a pie chart for the six factors shown in various colors. The merit of using the pie chart is that we could quickly understand the proportions for different factors at a glance. Factors with greater importance correspond to larger areas in the pie chart.

Figure 4.10(b) is a bar chart for the six factors. The bar chart provides a good view of the exact values of factors' correlation coefficients. However, instead of taking the absolute value as we did in the pie chart, we kept the sign for the coefficients such that

we could analyze the effect of factors both qualitative and quantitatively. Coefficients with positive signs represent that the corresponding factors have positive relationships between players' privacy preferences and vice versa. A positive relationship means that for a certain factor, based on the ranking criteria in Table 3.1, the higher the rank, the higher possibility that players are comfortable with the data collection or are more likely to allow the data collection. For example, in Figure 4.10(b) the factor **Location** is green and has the longest bar length. We could conclude that factor **Location** has the greatest impact on the players' privacy preferences, and the more space confidentiality the location has, the more comfortable they would feel. A counterexample is **Retention time**, whose correlation coefficient is negative, meaning that the shorter the time data is retained, the more comfortable players would feel.

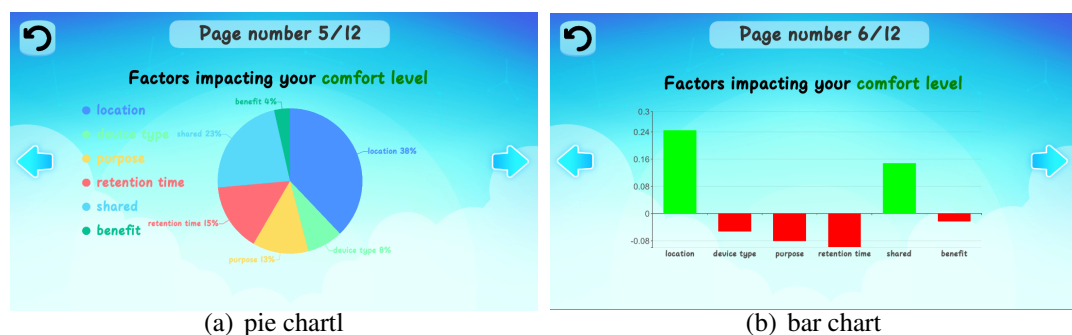


Figure 4.10: Visualize factors that impact comfort level

In the second part, we will provide feedback on pop-up quizzes, as shown in Figure 4.11. We will evaluate how players are familiar with the IoT devices based on their correctness on related devices, as shown in Figure 4.11(a). The usage of tips will also be presented in Figure 4.11(b).

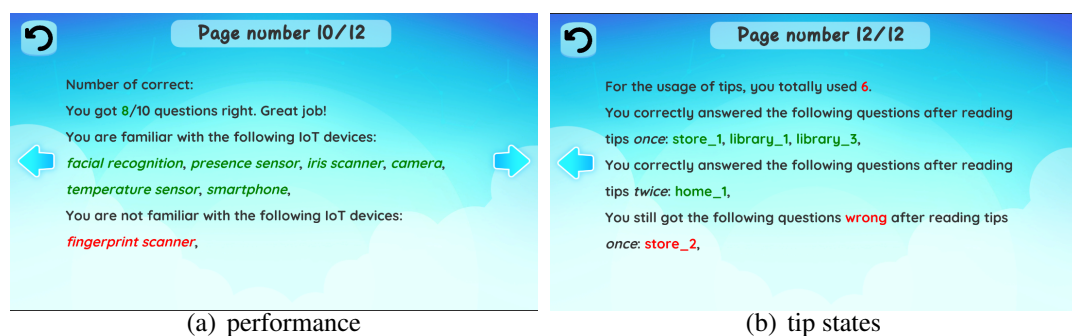


Figure 4.11: Players' performances on pop-up quizzes and tips states

Chapter 5

Game Implementation

In this chapter, we will introduce the detailed process of implementing the game, including the interfaces, button controller and statistical models. We will also discuss some techniques we used with Unity and C# scripts.

5.1 UGUI

Unity Graphical User Interface (UGUI) is a UI toolkit for developing user interfaces for games and applications provided by Unity in the version above 4.6. It is based on the UI system of `GameObject`, the base class for all entities in Unity scenes, to arrange and style user interfaces. It provides a functional event system module with various classes for mouse clicking, pointer moving, item dragging, and keyboard inputting. UGUI is widely used in this project to perform various functions, including interfaces, button controllers, and background.

5.1.1 Interface layouts

In the first step of implementation, we need to lay out the interfaces for every game scene. We bought a GUI kit from the internet containing game materials such as buttons, frames, icons, and fonts. Since UGUI uses `GameObject`-based UI system, we need to create a `GameObject` for each interface and adjust the corresponding parameters properly.

Figure 5.1 shows the inspection panel of a UGUI `GameObject`. It has three components: `Rect Transform`, `Canvas Renderer`, and `Image`. Almost every UGUI `GameObject` has the first two components, while the third component depends on the attribute

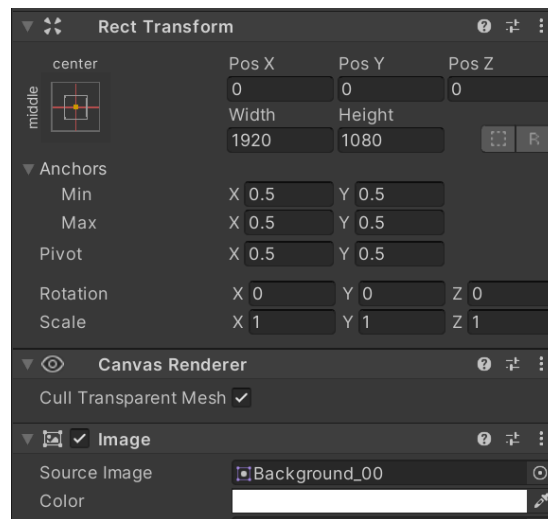


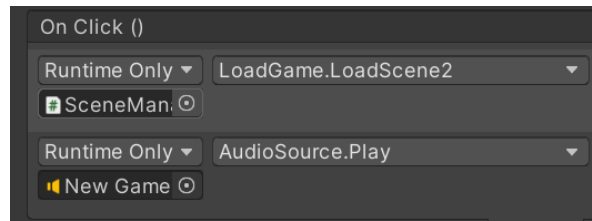
Figure 5.1: UGUI components

of the UGUI. For example, if we want to add a background to the scene, we need to set its source image in the Image component. On the other hand, if we want to add text to the dialogue box, it is apparent that we do not need a source image for text. Instead, we need a Text component that contains the plain texts we want to show in the dialogue box and other information such as the font size or the color of the text. The first component, Rect Transform, is the most crucial part of the UGUI because it controls the layout of interfaces. It could modify the interface's coordinate, size, anchors, pivot, and relative scale. In this example, the parameters in Figure 5.1 are used for the background, whose coordinate is set on original (0,0,0) and size is set to 1920×1080 , which is the resolution of our game. We repeated the same procedure for all interfaces.

5.1.2 Button controllers

Aside from the three components introduced in the previous section, there is another vital component in UGUI: Button. The Button component incorporates an event system trigger that handles the event when triggered by a pointer or keyboard. We added an `OnClick` event for every button such that all buttons could perform their duties. There are two ways to add `OnClick` event. The first one is directly adding events in the Button component, as shown in Figure 5.2, which is an example of the Button component for **New Game** button on the start page. We could see two clicking events attached to the **New Game** button. The first one calls the `LoadScene2()` method in the `SceneManager` class, as indicated in the dropdown window. The function will load a game scene specified by an index. The second event is an audio player. It calls the

`Play()` method in the `AudioSource` class, which plays a sound effect when the button is clicked.



(a) OnClick events in Button component



(b) New Game button in the start page

Figure 5.2: OnClick events in **New Game** button

Even though the `OnClick` events are easily deployed in the `Button` component, it also has some drawbacks. One of the main drawbacks is repeatability. Since button controllers are widely used in this game, setting events for every button controller would be tedious. Hence we use the other way to deploy `OnClick` events, the C# scripts. Unity has provided a class `IPointerClickHandler` which allows to set `OnClick` events with C# scripts. Based on the concept of Object Oriented Programming (OOP), we encapsulated a class `MyButton` inherited from the `Button` class which defines all basic events for buttons.

The class `Button` is included in the namespace `UnityEngine.UI` which defines every type of UGUI. By inheriting the `Button` class, we have all events for buttons. Furthermore, we redefined four events inherited from the `Button` class using the `override` keyword. The following four events are the methods we overrode: when the pointer is pressed, released, moved onto the UI, or moved out from the UI.

Except for the most commonly used buttons, such as the **New Game** in Figure 5.2, there are also some special buttons that have their specific duties. For instance, for those icon buttons such as department store and library shown in Figure 4.2, if players move a pointer onto the button, the size of the corresponding icon would be scaled by a factor of 1.2. As we did in the `MyButton` class, we encapsulated another class named `GeneralIconButton` for those icon buttons. We redefined the related events by overriding the methods inherited from its parents.

```
1 public class GeneralIconButton : MyButton
```

```
2 public class JumpSceneBtn : GeneralIconButton  
3 public class ReturnStageBtn : MyButton, IPointerClickHandler
```

Listing 5.1: Different child class inherited from Button

Listing 5.1 shows a list of child classes inherited from `Button`. Besides the first class introduced above, there are two more types of child classes: `JumpSceneBtn` inherited from `GeneralIconButton`, which provides the jumping scene function introduced in Section 4.2.4. `ReturnStageBtn` inherited from `MyButton`, which has an `OnClick` event transferring players to the main page from any other place in the game.

5.1.3 DOTween

DOTween is a powerful object-oriented animation engine for Unity. In this project, we use DOTween to make animations of UGUI, especially for UGUI transfer. In the interface of privacy questions shown in Figure 4.4, we added two buttons that allow the reference list on both sides of the dialogue box to gradually move up and down if players want to hide it. The static method `DOLocalMove()` will be called by the transform of a `GameObject` to change its position gradually.

```
1 void myCallback();  
2 Sequence seq = DOTween.Sequence();  
3 seq.AppendInterval(1.3f);  
4 seq.AppendCallback(myCallback);
```

Listing 5.2: DOTween sequence

We also used DOTween to delay function calls. The codes in Listing 5.2 show a way to use DOTween. Firstly, a `Sequence` object will be constructed. Then the method `AppendInterval()` will be called, with a float number as input. This method will tell the sequence how long it has to wait. The code in line 3 tells the sequence to wait for 1.3 seconds. The last step is to append the callback function, defined in `myCallback()`, to the sequence so that the callback function would be called after a waiting time. This technique is frequently used in the implementation of mini-game.

5.2 Main Content Implementation

The hierarchy panel on the left side of Figure 5.3 shows all scenes built in this game, including an introductory page, a main page, a shop, a result page, a mini-game, two tutorials, and six locations for different scenarios of privacy questions and pop-up quizzes.

In this section, we will introduce the implementation of the main content of this game. Specifically, we will introduce the detailed implementation procedures of the privacy questions and pop-up quizzes, mini-game, as well as the preference analysis models and performance feedback pages.

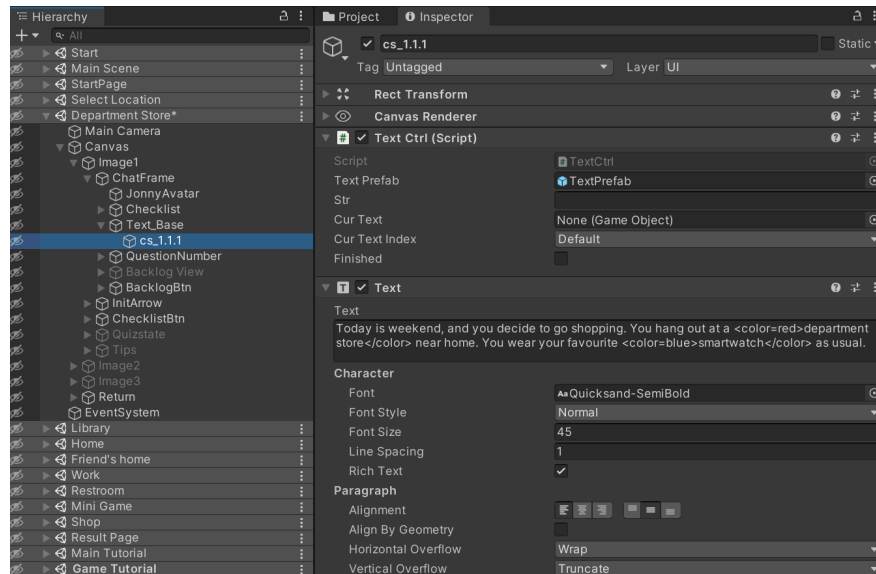


Figure 5.3: Game scenes and inspection panel for texts of privacy questions

5.2.1 Scenarios setup

The assets we set for each scenario are also in the hierarchy panel in Figure 5.3, which illustrates a half-expanded hierarchy of GameObjects in the department store scene. The three roots in the hierarchy from top to down are Main Camera, Canvas, and EventSystem. The Main Camera is a device through which the players view the world. Players would not see anything without it. The Canvas is the parent of every UGUI GameObject, without which players would not see any UGUI GameObject. The third one is EventSystem, an empty GameObject with a C# scripts for triggering events. Without it, all the system events, including button controls and keyboard inputs, would be invalid.

We put three images under the Canvas root to simulate three different scenes. Each image has its own child GameObjects, such as the dialogue box, jumping scene buttons, and a reference list. We use the dialogue box to communicate with players, telling them the setup of each scenario and asking them privacy questions. The right side of Figure 5.3 is an inspection panel of the dialogue box. The inspection panel shows all components that attached to the GameObject of dialogue box. Except for

the two compulsory components for UGUI described in Section 5.1.1, we also added a Text component for displaying texts and a TextCtrl class written in C# scripts for controlling texts. The TextCtrl class would firstly load a GameObject prefab from the resources folder. A GameObject Prefab in Unity could be used as a template to create new Prefab instances in the scene. The loaded TextPrefab has already well-defined the parameters in the Rect Transform component; hence, we do not have to adjust the parameters manually. Furthermore, the TextCtrl controls the behaviors of text GameObject, such as the display speed of text and the order of text display. We defined all texts in a string array encapsulated in a static class MainTextContent. The TextCtrl will read the texts that will be displayed in the dialogue box by indexing the string array in MainTextContent.

5.2.2 Player's information

We created a class Player that stores player's information in games. Some of the essential fields in Player are defined in Listing 5.3. Codes in lines 3 and 4 are variables for username and current points. Using generic programming skills, we could create a list of variables with any types in the generic interface List. The attribute tipStates is a list of TipState instances, which are class objects that record the state of tips, including the number of used tips and the corresponding quiz ID for which players have used tips. Similar to tipStates, the attribute quizstates is a list of QuizState instances. The class QuizState has four attributes: two string variables for the current quiz ID and the correct answer to that quiz, an integer for the number of tries players did on that quiz, and a boolean indicating whether players answered correct or not on that quiz. The attribute answers is a list of PrivacyQuestion instances, and each instance records players' choices for privacy questions. The PrivacyQuestion class has three attributes: a string variable for the current scenario ID and two double variables for comfort level and decision corresponding to that scenario. The static method Init() is used to initialize variables in Player.

```
1 public class Player
2 {
3     static public string userName;
4     static public float points;
5     static public List<TipState> tipStates;
6     static public List<QuizState> quizStates;
7     static public List<PrivacyQuestion> answers;
8     static public void Init();
```

9 }

Listing 5.3: Player class fields

When players make their choices of comfort levels and decisions, we need to update the information in the class `Player`. The information is updated as soon as players click the buttons for comfort levels and decisions, illustrated in Figure 4.5. Those buttons are attached with a `GeneralIconButton` component inherited from `MyButton`. Although they have some basic `OnClick` events, we also need to add a callback function that updates the player's information. Fortunately, the `Button` class has provided a method to dynamically add callback functions by calling `onClick.AddListener()`. The argument for `AddListener()` could be a function pointer or lambda expression.

Figure 5.4 shows an example of how we add a callback function to the “Allow” button for the decision on data collection. Firstly, we use `Find()` function to search the scenario ID that corresponds to the privacy question and set the decision to 1 for allowing and 0 for denying the data collection. After updating the information, we will save the data immediately by calling the method `SaveByJSON()`. Then we will push the text that tells players what choice they have made into the log. Subsequently, we will use the index of the current text in the dialogue box to instantiate the `GameObject` for the subsequent text in the string array in `MainTextContent` and finally destroy the current `GameObject`.

```
allow.GetComponent<GeneralIconButton>().onClick.AddListener(()=>
{
    Player.answers.Find(PrivacyQuestion => PrivacyQuestion.scenarioID == scenarioID).decision = 1;
    Save.SaveByJSON();
    Backlog.backlog.Add("You chose: <color=green>Allow data collection</color>");
    MainText(textIndex);
    GameObject.Destroy(decision);
});
```

Figure 5.4: Callback function for “allow” button

We will repeat the whole process mentioned above in the section on pop-up quizzes. The callback function for updating the information about players' answers for the quizzes and the state of used tips will be implemented in the icon buttons for IoT devices in Figure 4.6. We designed an algorithm to randomly sample the IoT devices from the device pool without replacement. The detailed implementation method for the random sampling algorithm uses two arrays, one representing the sampled devices and the other representing the index of devices yet to be sampled. We will add a callback function for the icon button of each sampled device.

5.2.3 Save game files

There are multiple ways to store players' information about their game progress. The simplest method is to use serialized field, yet it could only keep some simple variables such as integers, and it does not support complicated data structures. Hence we decided to use JSON files to save the game data. Figure 5.5 shows two static methods for saving and loading data, encapsulated into a class `Save`. In `SaveByJson()`, we firstly create a `playerinfo` instance, which contains all the player's information mentioned in Section 5.2.2. Then we call `JsonUtility.ToJson()` to convert player's information to JSON strings. In the following codes, a save file directory will be created if the directory does not exist, and then a stream writer object will be created for the system I/O. The last step is to write JSON strings to the stream writer and finally close the stream writer.

The procedure in function `LoadByJson()` is pretty much the same as that in the function `SaveByJson()` but in the opposite direction. In these steps, we will load the stored JSON strings to `playerinfo` such that players' game progressions are restored. It is worth noting that the comparison search for usernames mentioned in Section 4.2.1 was also implemented in the class `Save`. The procedure is similar to loading by JSON, in which we use a stream reader to get a list of players' data. Then we will do a string comparison with the usernames stored in the list.

5.2.4 Statistical models and personalized feedback

There will be two statistical models for individuals' privacy preferences, one for comfort levels and the other for decisions. The well-designed 75 scenarios are encoded into a `Dictionary` in class `ScenarioCode`. The codes in Listing 5.4 illustrate the declaration of the variable `scenarios` and an example of how we encoded a scenario into it. The key in the `Dictionary` is string, which represents the scenario ID, and the associated value is a list of doubles, which represent the value of different factors in the scenario. By referring to the code book in Table 3.1, the code in line 5 could be translated into the following texts:

In scenario ID 1, you are at a department store (0), and your smartphone (0.5) is keeping track of your specific position (0.5). Your position is used by the device to determine possible escape routes in the case of an emergency or a hazard (0.9). This data will be kept on your phone until you leave the shop (0). Your data will not be shared with other authorities (0) and you will be benefit from this data collection (1).

```

static public void SaveByJSON()
{
    // create a playerinfo object
    PlayerInfo playerInfo = CreateSave();
    // convert playerinfo to json string
    string JSONString = JsonUtility.ToJson(playerInfo, true);
    // create stream writer object
    if (!Directory.Exists(Application.dataPath + "/Save/"))
        Directory.CreateDirectory(Application.dataPath + "/Save/");
    StreamWriter sw = new StreamWriter(Application.dataPath + "/Save/playerInfo_" + Player.userName + ".json");
    // write json string to stream writer
    sw.Write(JSONString);
    // close the sw
    sw.Close();
}

```

(a) SaveByJson

```

static public PlayerInfo LoadByJSON(string userName)
{
    if (File.Exists(Application.dataPath + "/Save/playerInfo_" + userName + ".json"))
    {
        StreamReader sr = new StreamReader(Application.dataPath + "/Save/playerInfo_" + userName + ".json");
        string JsonString = sr.ReadToEnd();
        sr.Close();
        PlayerInfo playerInfo = JsonUtility.FromJson<PlayerInfo>(JsonString);
        return playerInfo;
    }
    else
        Debug.LogError("File Not Found.");
    return null;
}

```

(b) LoadByJson

Figure 5.5: Save and load functions

```

1 public class ScenarioCode
2 {
3     static public Dictionary<string, List<double>>> scenarios =
4     new Dictionary<string, List<double>>>{
5         {cs_ID_1, new List<double>{0, 0.5, 0.9, 0, 0, 1}},};
6 }

```

Listing 5.4: Scenario code example

We will use Math.NET Numerics to build our statistical models. Math.NET is an open source library providing algorithms and methods for numerical computations for C# and related .Net language just as SciPy for python. The codes to calculate Spearman's Rank Correlation Coefficient (SRCC) are shown in Listing 5.5. The argument of function `DenseOfArray()` in line 3 is the preprocessed dataset that includes encoded values for scenarios and players' choices for corresponding privacy questions. This function converts the whole dataset into a `Matrix` such that the statistic method `SpearmanMatrix` in line 4 could calculate the correlation coefficient matrix for the dataset. Subsequently, we will extract the columns for the correlation coefficients of comfort level and decision and put them into the pie chart and the bar chart in the results pages, as illustrated in Figure 4.10. Furthermore, we will also take the maximum absolute values of coefficients by calling `Math.abs()` to calculate the most and lest

important factors that affect individuals' privacy preferences.

```

1 using MathNet.Numerics.Statistics;
2 public class PreferenceModel : MonoBehaviour
3 {
4     void Start() {
5         Matrix<double> matrix = Matrix<double>.Build.DenseOfArray(data);
6         Matrix<double> coeff = Correlation.SpearmanMatrix(matrix);
7     }

```

Listing 5.5: Calculate correlation coefficient with MathNet

The implementations of personalized pop-up quiz feedback are pretty straightforward. Firstly, we will iterate instances in `tipsStates` and `quizStates` to count the number of correct answers grouped by the type of IoT devices and calculate the accuracy for each device. Then we will rank the accuracy in ascending order to tell players which IoT devices they are familiar with and which ones they aren't. Regarding the usage of tips, we will report how many tips players used on each quiz. As we introduced in Section 4.2.3, there are two levels of tips for each quiz. We will notify players which quizzes they did correct with the help of tips' first level or second level. The use of the first level of tips indicates that players have some understanding of the IoT device. Still, they are not familiar enough with it. Using the second level of tips means that players only have some limited knowledge about the IoT device.

5.3 Mini-Game

The implementations of the mini-game are complicated, and due to page limits, we will not discuss the procedure at a detailed level. Instead, we will briefly introduce some techniques we used in the mini-game.

5.3.1 Delegate

Delegate is a programming grammar in C# references to methods with a particular parameter list and return type. It behaves like a function pointer in C++ but it is fully object-oriented, which encapsulates both object instances and methods. In the implementation of the mini-game, we used delegates to define the behaviors of each block in the game board, managed by the class `EventDispatcher`.

```

1 public class EventDispatcher
2 {

```

```
3     public delegate void MyEventHandler(params object[] objs);  
4     public void Regist(string eventName, MyEventHandler handler);  
5     public void Dispatch(string eventName, params object[] objs);  
6 }
```

Listing 5.6: Event dispatcher

The codes in Listing 5.6 are some of the essential fields in the class `EventDispatcher`. In line 3, we used the keyword `delegate` to qualify the function object `MyEventHandler` which defines the behavior of each block. As we introduced in Section 4.2.5, blocks are eliminated when more than three lie in a line. Hence we need to add an event `EVENT_DISAPPEAR` that destroys the `GameObject` relating to the eliminated, and another event `EVENT_ADD_POINTS` that adds points to players' current points for eliminating blocks. There are more events attached to the block, such as the event when a mouse drags the block or when players use tools.

We will use the method `Regist()` to register events we defined in different classes. This method serves as an event pool. When an `EventDispatcher` instance is created, we call this method to register events to the event pool by passing the event name and corresponding function object to the method. Afterward, we call the method `Dispatch` in line 5 to dispatch the event.

5.3.2 Object pooling

Considering that the mini-game involves frequent `GameObject` instantiating and destroying, an intelligent way to implement it is using object pooling. Object pooling helps us optimize the program and reduce CPU burdens when we need to create and destroy `GameObjects` rapidly. The design pattern in object Pooling is that we pre-instantiate all the objects we need ahead of time and simply activate the `GameObjects` required during the gameplay. After the `GameObjects` are used, we will not destroy them. Instead, we will inactive and recycle them to the object pool for subsequent use.

In the class `EffectSpawner`, we declared a object pool `myPool` whose data structure is a `GameObject` queue. When we want to use a `GameObject`, we first check the number of items in the queue. If there is more than one item in the queue, we simply take one item by calling `dequeue`. Otherwise, we would instantiate one `GameObject` using the prefab and then active the `GameObject` in the hierarchy. Additionally, there is a callback function to deactivate the `GameObject` and push it into the queue when we are done with the `GameObject`.

Chapter 6

Evaluation and User Study

In this chapter, we will talk about the evaluation and user study of this project. The game will be evaluated based on a game experience questionnaire and the points earned on the System Usability Scale (SUS).

6.1 User Study Process

To be eligible to conduct the user study, we carefully write a personal information sheet (PIS), attached in Appendix, to get consent from participants to play the game and do the game experience survey for the user study. We applied for an ethics application, with application number 60811, and got approved according to the Informatics Research Ethics Process.

We uploaded the PIS to the OneDrive cloud, and a hyperlink generated by the PIS would be shared with potential participants. If they are agreed to the PIS, we will share a hyperlink for the game, and participants need to download the game to their local disks. A “README” file is attached with the game, which they need to read ahead of playing the game. When participants finish playing the game, they will use the link in the “README” file to open an online survey created by Microsoft form. There is an optional step that participants could choose to decide whether share the data file which is generated during the game with us for further study. We strictly follow the rules in ethics, and all data will be shared anonymously and securely stored in University’s secure encrypted cloud storage services.

6.2 Game Experience Survey

We have recruited seven participants who agreed to take part in this study. They followed the user study processes and filled out a game experience questionnaire. The complete questionnaire form is appended in the Appendix. This questionnaire is used to evaluate the system usability, as well as to collect feedback about each functionality of the game.

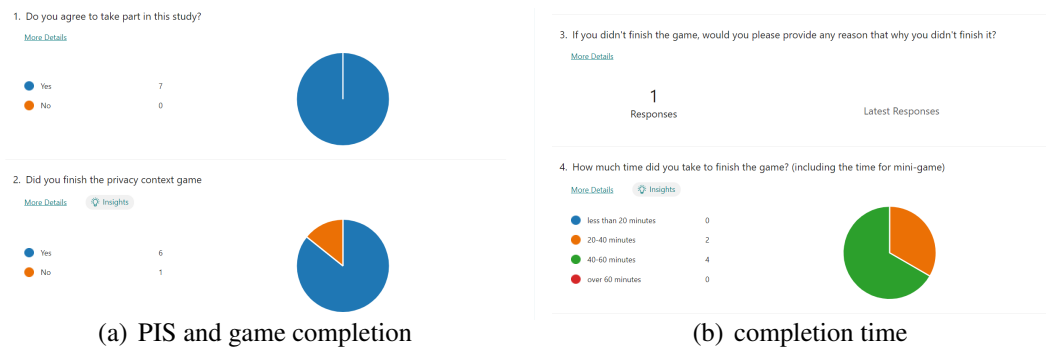


Figure 6.1: Game experience survey part 1

The survey has a total of 16 questions with different branches. In Figure 6.1, we could find that all seven participants agreed to participate in this study. However, one participant did not finish our game, and the reason was provided in question 3: there were too many questions in this game, which cost time beyond expectation. The overall completion time for the other six participants, including the mini-game, is shown in Figure 6.1(b). All of them completed the game within one hour.

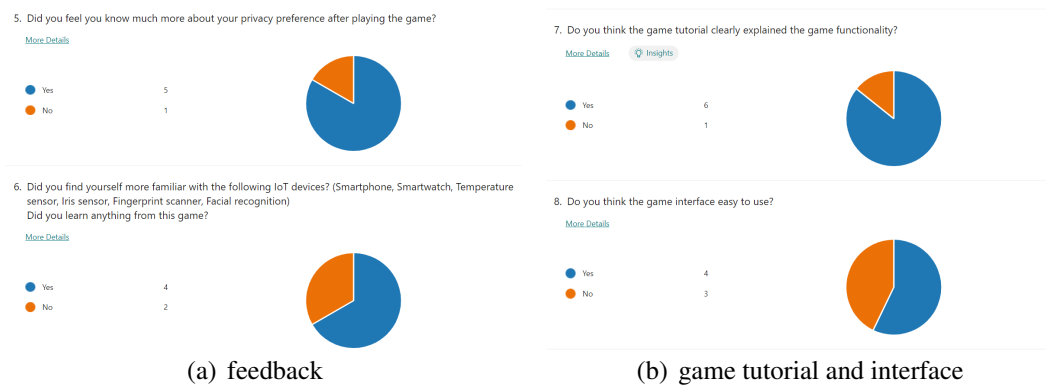


Figure 6.2: Game experience survey part 2

For participants who completed the game, we asked them two questions regarding two project aims introduced in Section 1.2, as shown in Figure 6.2(a). Almost all

participants feel they learned more about their privacy preferences, and two-thirds feel they are more familiar with the IoT devices involved in the game. Questions 7 and 8 in Figure 6.2(b) aim to evaluate the clarity of the game tutorial and interface. The result indicates that we did well on the game tutorial, but the game interface remains to be improved.

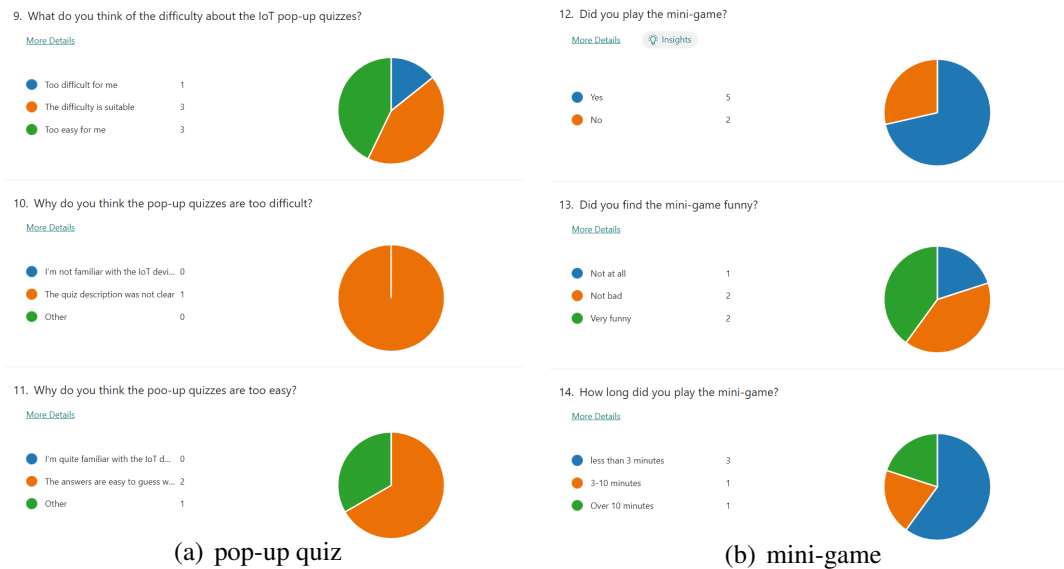


Figure 6.3: Game experience survey part 3

Figure 6.3(a) and 6.3(b) are questions for evaluating the pop-up quizzes and the mini-game. All participants took the quizzes, and nearly half thought the quizzes were too easy. According to their feedback, it is partially due to our statements for the quizzes. The hints we gave players are so obvious that they could guess the correct answers without prior knowledge about the IoT devices. Nevertheless, our primary target is to educate players rather than make things difficult. For the evaluation of the mini-game, five participants did try it on, and their evaluations were that the mini-game was better than average. It is worth noting that one participant enjoyed the mini-game for more than ten minutes. It represents that this mini-game has good playability.

The last part of the game experience survey is a system usability scale (SUS). The SUS has been a quick method to evaluate the usability of different human-agent systems and software for almost 20 years [22]. We received five evaluations from participants for the SUS. We computed the SUS score using the specified framework for our system and achieved a score of 71.6 out of 100, representing that our system is better than the average, whose score is 68 [4]. According to Figure 6.4, some participants considered the system unnecessarily complex but thought it was easy to use. They

16. System Usability Scale

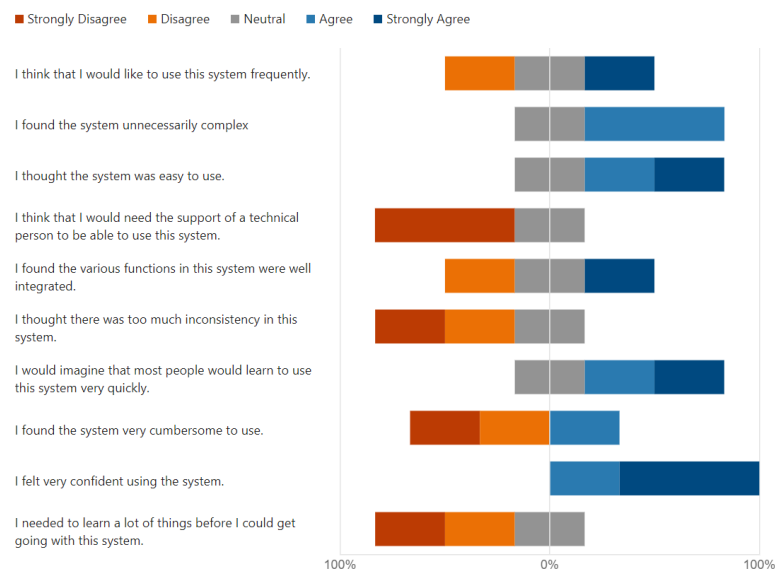
[More Details](#)

Figure 6.4: System Usability Scale (SUS)

also believed that most people would quickly learn to use this system. In the aspect of functionality, we integrated various functions in this system well and consistently.

6.3 Further Study

Participants who have completed the game also uploaded their saved data files to the OneDrive cloud as indicated in the instructions. Subsequently, we downloaded their data files and did some analysis. We have some interesting findings and we will briefly summarize the results.

- For most participants, the importance of factors affecting their comfort levels and that of factors affecting decisions are highly consistent, which means if they are more comfortable with the data collection, they are more likely to approve that collection.

Table 6.1 shows the importance of factors for one anonymous participant. The second and third columns are the correlational coefficients of factors that impact an individual's comfort level and decision, respectively. Each row represents the "coefficient pair" of a factor. For example, the coefficient pair of **Location** is -0.015 for comfort level and -0.008 for decision. According to a qualitative analysis of these coefficients, we could find that the sign of each coefficient pair is consistent. Still taking

Factor name	Comfort level	Decision
Location	-0.015	-0.008
Device type	0.024	0.035
Purpose	0.001	0.085
Retention time	-0.087	-0.114
Shared	0.09	0.066
Benefit	-0.035	-0.046

Table 6.1: Correlational coefficients of different factors for one anonymous participant

the **Location** factor as an example, the negative sign on both comfort level and the decision means that the more space confidentiality, the less comfortable they would feel, and the less likely they would approve the data collection. The rules also apply to all other factors. The finding on this consistency is exciting because it would help to simplify individuals' privacy preferences. When they face more complicated scenarios in future, they will be more confident in making decisions on data collection.

- Some participants changed their minds in two consecutive scenarios in which only one factor changed.

According to an anonymous participant's data, their choices changed in two consecutive scenarios, as indicated in Figure 6.5. In scenario ID `1b_ID_2`, we told the participant that an iris scanner would collect their biometric information in the library, and the data will be kept forever. The participant's associated comfort level was `Uncomfortable(2)` and the decision was `Deny(0)`. Later, in scenario ID `1b_ID_3` we told that the data would be kept until the purpose was satisfied, and the comfort level increased by one, and the decision became `Allow(1)`. The same thing happened in several consecutive scenarios. According to our preliminary speculation, we thought these were not coincident. We speculate that it is conducive to making decisions on privacy questions when two similar scenarios are compared with only one factor changed.

```
"scenarioID": "1b_ID_2", "comfortable": 2.0, "decision": 0.0
"scenarioID": "1b_ID_3", "comfortable": 3.0, "decision": 1.0
```

Figure 6.5: Choices on two consecutive scenario for one anonymous participant

Chapter 7

Conclusion

In this project, we used the Unity engine to build an educational game that helps players learn about their privacy preferences and become more familiar with IoT devices. We firstly designed 75 scenarios of data collection in which players would face in the game and make their choices on comfort levels and decisions regarding different scenarios. To incorporate education with entertainment, we designed a pop-up quizzes system with tips and embedded a mini-game into our project to motivate players to achieve higher grades on the quizzes. The players' information were stored in JSON files such that their game sessions could be restored. We also built two statistical models for analyzing the importance of factors by calculating correlation coefficients. We displayed the analyzing results on both pop-up quizzes and privacy questions in the results pages. A user study was conducted after the implementation of the game to evaluate the game and the system's usability. We received good grades on the System Usability Scale, and achieved two primary project aims. We also studied saved game files uploaded by participants and had exciting findings on their privacy preferences.

There are still some limitations of this project. According to the game experience survey feedback, the difficulty of pop-up quizzes was too easy, so players might lose motivation to study in the game. Hence we should concentrate more on designing the questions. Secondly, we noticed that a participant quit the game without finishing the privacy questions. Too many privacy questions may reduce their interest. We must pay more attention to players' experiences when designing the game. Although there are still limitations to the project, and many aspects remain to be improved, we gained experience through this project. In summation, we successfully achieved our tasks, and we believe this project will profoundly impact us in future research on individuals' privacy preferences and developing educational games.

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Appendix A

Participant Information Sheet

Participant Information Sheet

Project title:	Personalized educational game in human-agent systems
Principal investigator:	Nadin Kokciyan
Researcher collecting data:	Yibin Xia
Funder (if applicable):	None

This study was certified according to the Informatics Research Ethics Process, RT number 60811. Please take time to read the following information carefully. You should keep this page for your records.

Who are the researchers?

The lead researcher is Yibin Xia (s2196789@ed.ac.uk) and the supervisor is Nadin Kokciyan (nadin.kokciyan@ed.ac.uk).

What is the purpose of the study?

We developed an educational game using Unity to investigate people's privacy preferences when they are interacting with IoT devices (e.g., camera, smartwatch, smartphone) in different scenarios. Those scenarios can have different effect on people's attitudes (e.g., comfortable, uncomfortable) and decisions (e.g., approve or reject) towards data collections of IoT devices. In this study, we aim to find the most important factors that can influence those preferences by having the participants to play with the game and providing feedback. The study aims also to help participants acquire more knowledge about the functions of different IoT devices by giving participants a few quizzes and explanations in the form of second person narration.

Why have I been asked to take part?

We are looking for participants who have experience with college education, preferably a University of Edinburgh student with computer science major or taking relevant courses. We believe that you are capable of using or have experience with various kinds of IoT devices and could provide valuable feedback for our study.

Do I have to take part?

No – participation in this study is entirely up to you. You can withdraw from the study at any time, without giving a reason. After this point, personal data will be deleted and anonymised data will be combined such that it is impossible to remove individual information from the analysis. Your rights will not be affected. If you wish to withdraw, contact the PI. We will keep copies of your original consent, and of your withdrawal request.

What will happen if I decide to take part?

If you decide to take part, you will receive a link that you can download our game. Then you can follow the instructions we provided to play the game. In the game, firstly you will be asked to give a name. It doesn't need to be a real name, and any characters between length of 3 to 10 are fine. Then your account will be created and saved automatically. You will experience totally 75 different privacy scenarios by selecting 5 different locations. For each scenario, you will be asked questions about whether you are comfortable with and whether you allow the data collection in this scenario. Those preferences for each scenario will be collected and analysed in the study. There is also a mini-game embedded in this game you can play with and the score for the mini game will be recorded during the game. The duration will be approximately 20-30 minutes for the privacy question and as long as you want for the mini-game. After you finish the game, you may be asked to complete a questionnaire about your experience with the game. You can come back anytime by clicking the "continue" button in the start page of the game and loading the data with your name. The preference analysing results will be displayed when you finish the game.

Are there any risks associated with taking part?

There are no significant risks associated with participation.

Are there any benefits associated with taking part?

You may acquire knowledge about various kinds of IoT devices and have better understanding about your privacy preference when it comes to making privacy decisions.

What will happen to the results of this study?



The results of this study may be summarised in published articles, reports and presentations. Quotes or key findings will be anonymized: We will remove any information that could, in our assessment, allow anyone to identify you. With your consent, information can also be used for future research. Your data may be archived for a maximum of 1 year. All potentially identifiable data will be deleted within this timeframe if it has not already been deleted as part of anonymization.

Data protection and confidentiality.

Your data will be processed in accordance with Data Protection Law. All information collected about you will be kept strictly confidential. Your data will be referred to by a unique participant number rather than by name. Your data will only be viewed by the researcher/research team Yibin Xia, and his supervisor Nadin Kokciyan.

All electronic data will be stored on a password-protected encrypted computer, on the School of Informatics' secure file servers, or on the University's secure encrypted cloud storage services (DataShare, ownCloud, or Sharepoint) and all paper records will be stored in a locked filing cabinet in the PI's office. Your consent information will be kept separately from your responses in order to minimise risk.

What are my data protection rights?

The University of Edinburgh is a Data Controller for the information you provide. You have the right to access information held about you. Your right of access can be exercised in accordance Data Protection Law. You also have other rights including rights of correction, erasure and objection. For more details, including the right to lodge a complaint with the Information Commissioner's Office, please visit www.ico.org.uk. Questions, comments and requests about your personal data can also be sent to the University Data Protection Officer at dpo@ed.ac.uk.

Who can I contact?

If you have any further questions about the study, please contact the lead researcher, Yibin Xia <s2196789@ed.ac.uk>.

If you wish to make a complaint about the study, please contact inf-ethics@inf.ed.ac.uk. When you contact us, please provide the study title and detail the nature of your complaint.



Updated information.

If the research project changes in any way, an updated Participant Information Sheet will be made available on <http://web.inf.ed.ac.uk/infweb/research/study-updates>.

General information.

For general information about how we use your data, go to: edin.ac/privacy-research



Appendix B

Game Experience Questionnaire

Branching options

1. Do you agree to take part in this study? *

☐ Yes

Go to 2. Did you finish the ...

☐ No

Go to End of the form

2. Did you finish the privacy context game

☐ Yes

Go to 4. How much time did you ta...

☐ No

Go to 3. If you didn't finish the ga...

3. If you didn't finish the game, would you please provide any reason that why you didn't finish it?

Go to 8. Do you think the game inte...

4. How much time did you take to finish the game? (including the time for mini-game)

- ☐ less than 20 minutes
- ☐ 20-40 minutes
- ☐ 40-60 minutes
- ☐ over 60 minutes

5. Did you feel you know much more about your privacy preference after playing the game?

- ☐ Yes
- ☐ No

6. Did you find yourself more familiar with the following IoT devices?
(Smartphone, Smartwatch, Temperature sensor, Iris sensor, Fingerprint scanner, Facial recognition)

Did you learn anything from this game?

- ☐ Yes
- ☐ No

7. Do you think the game tutorial clearly explained the game functionality?

- ☐ Yes
- ☐ No

8. Do you think the game interface easy to use?

- ☐ Yes
- ☐ No

9. What do you think of the difficulty about the IoT pop-up quizzes?

- ☐ Too difficult for me
- ☐ The difficulty is suitable
- ☐ Too easy for me

Go to [10. Why do you think the po...](#)

Go to [12. Did you play the mini-ga...](#)

Go to [11. Why do you think the po...](#)

10. Why do you think the pop-up quizzes are too difficult?

- ☐ I'm not familiar with the IoT devices
- ☐ The quiz description was not clear
- ☐ Other

11. Why do you think the poo-up quizzes are too easy?

- ☐ I'm quite familiar with the IoT devices
- ☐ The answers are easy to guess with out any prior knowledge
- ☐ Other

12. Did you play the mini-game?

- ☐ Yes
- ☐ No

13. Did you find the mini-game funny?

- ☐ Not at all
- ☐ Not bad
- ☐ Very funny

14. How long did you play the mini-game?

- ☐ less than 3 minutes
- ☐ 3-10 minutes
- ☐ Over 10 minutes

15. Could you please provide any other comments regarding your experience with this game?

16. System Usability Scale

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
I found the system unnecessarily complex	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

I thought
there was too
much
inconsistency
in this
system.

☐

☐

☐

☐

☐

I would
imagine that
most people
would learn
to use this
system very
quickly.

☐

☐

☐

☐

☐

I found the
system very
cumbersome
to use.

☐

☐

☐

☐

☐

I felt very
confident
using the
system.

☐

☐

☐

☐

☐

I needed to
learn a lot of
things before
I could get
going with
this system.

☐

☐

☐

☐

☐

Go to [Next](#)

